

ARTIFICIAL NEURAL NETWORKS (ANN)

Topics: Linear Regression • Logistic Regression • Activation Functions • Loss Functions • Perceptron • NumPy • Pandas

Instructions

- Section A: contains Multiple Select Questions (MSQs). One or more options may be correct. Select all correct options.
- Section B: Mark each statement as True (T) or False (F).

Section A — Multiple Select Questions (MSQ)

Select all options that are correct. Each question may have one or more correct answers.

- Q1.** Consider the model $\hat{y} = w_1x_1 + w_2x_2 + b$. Which statements are correct?
- A. The model is linear with respect to its parameters.
 - B. The model can represent a nonlinear decision boundary in the original feature space.
 - C. Changing b shifts the prediction without changing the contribution of x_1 and x_2 .
 - D. If $w_1 = w_2 = 0$, the prediction is always b .
 - E. The model necessarily requires an activation function.
- Q2.** For a neuron $z = w_1x_1 + w_2x_2 + b$, which statements are correct?
- A. Increasing w_1 always increases z .
 - B. Increasing w_1 increases z only when $x_1 > 0$.
 - C. Increasing b by 1 increases z by 1 regardless of the input values.
 - D. If $x_1 = 0$, changing w_1 has no effect on z .
 - E. The bias controls the slope of the decision boundary.
- Q3.** For logistic regression, $\hat{y} = \sigma(w^T x + b)$, which statements are correct?
- A. The output is constrained between 0 and 1.
 - B. The sigmoid makes the decision boundary nonlinear in the original input space.
 - C. The underlying decision boundary $w^T x + b = 0$ is linear.
 - D. Logistic regression can be interpreted as estimating a probability for binary classification.
 - E. Logistic regression cannot be optimized using gradient-based methods.
- Q4.** Consider $\text{ReLU}(z) = \max(0, z)$. Which statements are correct?
- A. ReLU produces zero output for all negative inputs.
 - B. ReLU has a constant derivative of 1 for positive inputs.
 - C. ReLU is nonlinear.
 - D. ReLU can suffer from the dying ReLU problem.
 - E. ReLU always produces outputs between 0 and 1.
- Q5.** Consider a neural network consisting of several layers, but without any nonlinear activation function. Which statements are correct?
- A. Multiple linear layers can be mathematically reduced to a single linear transformation.
 - B. Increasing the number of linear layers automatically makes the model nonlinear.
 - C. Nonlinear activation functions allow neural networks to model nonlinear relationships.
 - D. A deep network without nonlinear activations has limited expressive power compared with a network containing nonlinear activations.
 - E. Bias terms alone make a multilayer network nonlinear.
- Q6.** Suppose $h = W_1x + b_1$ and $y = W_2h + b_2$. Which statements are correct?
- A. The complete system can be represented as a single affine transformation of x .
 - B. The composition is nonlinear because there are two weight matrices.
 - C. The resulting transformation can be written as $y = W_2W_1x + W_2b_1 + b_2$.
 - D. Adding ReLU between the two layers changes this property.
 - E. Increasing the number of such purely linear layers necessarily increases the representational capacity.
- Q7.** For a binary perceptron with $\hat{y} = 1$ if $w^T x + b \geq 0$ and 0 otherwise, which statements are correct?
- A. A perceptron can learn linearly separable binary classification problems.
 - B. A single perceptron can solve the XOR problem.
 - C. The decision boundary is a hyperplane.
 - D. The bias can shift the decision boundary.
 - E. The perceptron fundamentally requires a sigmoid activation function.
- Q8.** Why cannot a single-layer perceptron correctly classify the standard XOR problem?
- A. XOR is not linearly separable.
 - B. A single perceptron produces a linear decision boundary.
 - C. Adding more training examples automatically makes XOR linearly separable.
 - D. A multilayer network with nonlinear activations can represent XOR.
 - E. Increasing the learning rate guarantees that a single perceptron will solve XOR.
- Q9.** For $\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$, which statements are correct?
- A. Large errors receive disproportionately larger penalties than small errors.
 - B. MSE is always non-negative.
 - C. MSE becomes zero when all predictions exactly

match their targets.

- D.** MSE is generally appropriate for regression problems.
- E.** MSE is always the best loss function for binary classification.
- Q10.** For binary cross-entropy $L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$, which statements are correct?
- A.** For $y = 1$, the loss is $-\log(\hat{y})$.
- B.** Predicting a probability very close to 0 for a true class of 1 results in a large loss.
- C.** Predicting 0.5 always gives zero loss.
- D.** Binary cross-entropy strongly penalizes confident incorrect predictions.
- E.** The loss is minimized when the predicted probability matches the true class.
- Q11.** Consider $\mathbf{X}.\text{shape} = (100, 4)$, $\mathbf{w}.\text{shape} = (4,)$, and scalar b . For $z = \mathbf{X}@\mathbf{w} + b$, which statements are correct?
- A.** z has shape $(100,)$.
- B.** Each of the 100 rows of X produces one value of z .
- C.** w represents four weights corresponding to the four features.
- D.** Matrix multiplication can replace an explicit Python loop over the 100 samples in this calculation.
- E.** $\mathbf{X}@\mathbf{w}$ is element-wise multiplication.
- Q12.** A dataset contains Age, Salary, Experience, and Purchased. Which operations are appropriate before training a machine-learning model?
- A.** Separate input features from the target variable.
- B.** Check for missing values.
- C.** Inspect data types and unusual values.
- D.** Randomly change feature values to make the dataset more balanced.
- E.** Check the distribution of numerical features.
- Q13.** Consider $z_1 = W_1x + b_1$, $a_1 = f(z_1)$, $z_2 = W_2a_1 + b_2$, and $\hat{y} = g(z_2)$. Which statements are correct?
- A.** z_1 is the pre-activation value of the first layer.
- B.** a_1 is obtained after applying f .
- C.** W_2 operates on a_1 , not directly on the original input x .
- D.** The forward pass calculates predictions from input data using the current parameters.
- E.** Backpropagation is required to calculate the forward-pass prediction.
- Q14.** A neural network has 4 input features, 5 hidden neurons, and 2 output neurons. Every neuron has one bias. Which statements are correct?
- A.** The first layer contains 20 weights.
- B.** The first layer contains 25 trainable parameters including biases.
- C.** The output layer contains 10 weights.
- D.** The output layer contains 12 trainable parameters including biases.
- E.** The complete network contains 37 trainable parameters.
- Q15.** For $z_1 = W_1x + b_1$, $a_1 = \text{ReLU}(z_1)$, $z_2 = W_2a_1 + b_2$, $\hat{y} = \sigma(z_2)$, with binary cross-entropy loss, which statements are correct?
- A.** W_1 determines how the original features are linearly combined before ReLU.
- B.** ReLU introduces nonlinearity into the network.
- C.** The sigmoid constrains the final output to $(0, 1)$.
- D.** Binary cross-entropy can penalize a highly confident incorrect prediction strongly.
- E.** Removing ReLU while keeping all other operations makes the entire network nonlinear because sigmoid is still present.
- F.** The parameters W_1, W_2, b_1, b_2 can be updated using gradients obtained through backpropagation.
- Q16.** Consider $h = W_1x + b_1$, $a_1 = \text{ReLU}(h)$, and $\hat{y} = W_2a_1 + b_2$. Suppose ReLU is removed. Which statements are correct?
- A.** The resulting network becomes an affine transformation of the input.
- B.** Increasing the number of hidden layers without nonlinear activations necessarily makes the network more expressive.
- C.** The composition of multiple linear/affine transformations can be represented by a single affine transformation.
- D.** Removing ReLU can significantly reduce the ability of the network to learn nonlinear relationships.
- E.** The bias terms alone are sufficient to make the network nonlinear.
- Q17.** Consider a sigmoid neuron $a = \sigma(z)$ with derivative $\sigma(z)(1 - \sigma(z))$. Suppose $z = 10$. Which statements are correct?
- A.** The sigmoid output is very close to 1.
- B.** The derivative of sigmoid is very small.
- C.** A small gradient can result in very small parameter updates during gradient descent.
- D.** This behavior can contribute to the vanishing-gradient problem in deep networks.
- E.** The derivative is largest when z is very large and positive.
- Q18.** A dataset is split into training and testing sets, but a scaler is fitted on the entire dataset before the split. Which statements are correct?
- A.** Information from the test set can influence the scaling parameters.
- B.** This can result in data leakage.
- C.** The reported test performance may become overly optimistic.
- D.** The correct approach is to fit the scaler using only the training data.
- E.** The test data should never be transformed using the scaler fitted on the training data.
- Q19.** A batch contains 32 samples with 6 features. The network is $\text{Dense}(10) \rightarrow \text{Dense}(5) \rightarrow \text{Dense}(1)$, with biases. Which statements are correct?
- A.** The input matrix has shape $(32, 6)$.
- B.** The weight matrix connecting the input to the first

hidden layer has shape (6, 10).

- C. The output of the first hidden layer has shape (32, 10).
- D. The weight matrix connecting the first hidden layer to the second hidden layer has shape (10, 5).
- E. The final output has shape (32, 1).
- F. The total number of trainable parameters is 121.
- Q20.** Consider Model A: $\hat{y} = \text{step}(w^T x + b)$; Model B: $\hat{y} = \sigma(w^T x + b)$; Model C: $h = \text{ReLU}(W_1 x + b_1)$, $\hat{y} = \sigma(W_2 h + b_2)$. Which statements are correct?
- A. Model A and Model B have a linear decision boundary when a threshold of 0.5 is used.
- B. Model C can represent nonlinear decision boundaries.
- C. Model B can output a probability-like value between 0 and 1.
- D. Model A and Model B necessarily have the same training behavior because they have the same $w^T x + b$.
- E. Model C can learn representations of the input before making the final classification.
- F. Replacing ReLU in Model C with no activation necessarily makes Model C equivalent to Model A.

Section B — Conceptual True/False

Mark each statement as True (T) or False (F). Students should be prepared to justify their answers.

1. A linear regression model can contain multiple input features and still be called a linear model, even though its output may change nonlinearly with respect to some transformed features.
2. Adding a bias term to a linear regression model increases the model's ability to shift the prediction but does not by itself introduce nonlinearity.
3. If two linear layers are stacked without an activation function between them, the resulting network can represent any nonlinear function provided that enough neurons are added.
4. A polynomial regression model of degree 2 is nonlinear with respect to its input variables but can still be linear with respect to its trainable coefficients.
5. If the training MSE of a regression model is zero, the model must also have zero error on unseen test data.
6. Logistic regression uses a nonlinear sigmoid function, but its binary decision boundary remains linear with respect to the original input features.
7. If two logistic regression models have identical weights but different bias values, their decision boundaries must have the same position.
8. A logistic regression model can produce a probability of 0.8 for a sample even though the corresponding class prediction after thresholding at 0.5 is class 1.
9. Replacing the sigmoid function in logistic regression with a step function does not change the mathematical optimization problem because both functions produce binary classifications.
10. Logistic regression can perfectly classify every possible binary dataset if sufficient training iterations are used.
11. A network containing only linear transformations cannot approximate a nonlinear function merely by increasing its depth.
12. ReLU introduces nonlinearity even though its mathematical expression is linear for positive inputs.
13. For a sigmoid neuron, increasing the magnitude of its input z indefinitely always results in a proportionally large increase in its output.
14. A sigmoid neuron can have a very small gradient even when its prediction is highly confident.
15. Replacing ReLU with another nonlinear activation function necessarily changes the type of decision boundary that the network can represent.
16. A single perceptron can represent any classification problem as long as its weights are optimized correctly.
17. The inability of a single perceptron to solve XOR is fundamentally related to linear separability rather than the number of training iterations.
18. Changing only the bias of a perceptron can change the location of its decision boundary without changing its orientation.
19. A perceptron with a step activation can be viewed as performing a threshold operation on a weighted combination of its inputs.
20. A lower loss value always means that the model will have better generalization performance on unseen data.
21. MSE penalizes an error of magnitude 4 more heavily than an error of magnitude 2 because the squared errors are 16 and 4 respectively.
22. In binary cross-entropy, predicting a probability of 0.99 for a true class of 0 is penalized more strongly than predicting 0.60 for the same sample.
23. If the gradient of the loss with respect to a particular weight is zero, that weight will not change during a standard gradient-descent update, assuming no other optimization mechanism modifies it.
24. Increasing the learning rate always decreases the number of iterations required for a neural network to reach its minimum loss.